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Original Article

Every other day stereotactic radiation therapy for the treatment of uveal melanoma decreases toxicity

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ABSTRACT

Background and purpose: To report the long-term results of stereotactic radiosurgery and fractionated stereotactic radiation therapy (SRS/FSRT) in patients with uveal melanoma (UM).
Materials and methods: We retrospectively evaluated the results of patients treated between 2007 and 2019. The primary endpoints were local control (LC), local recurrence-free survival (LRFS), enucleation-free survival (EFS) and treatment toxicity.
Results: 443 patients with 445 UMs were treated via CyberKnife®. According to the COMS classification, 70% of the tumors were small/medium and 30% were large. Median total RT dose was 54 Gy, median BED10 was 151 Gy. After a median 74-months follow-up, SRS/FSRT yielded an 83% overall LC rate. The 5- and 10-year LRFS rate was 74% and 56%, respectively. Patient age and the COMS size were prognostic for all survival endpoints. An increased SRS/FSRT dose was associated with higher LRFS and EFS rates. SRS/FSRT-related toxicity was observed in 49% of the eyes. Median visual acuity (VA) significantly deteriorated after SRS/FSRT in 76% of the treated eyes. The overall eye preservation rate was 62%, and the 5- and 10-year EFS rate was 64% and 36%, respectively. The delivery of FSRT every other day resulted in a significantly lower rate of toxicity and enucleation compared to FSRT on consecutive days.
Conclusion: A total dose of $\geq$45 Gy and $BED_{10Gy} \geq 112.5$ SRS/FSRT is associated with a higher LC rate in patients with UM. Despite the favorable outcomes, treatment toxicity is the major limitation of this treatment. Toxicity and enucleation can be minimized by treating the eye every other day.

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults [1]. The management of UM is directed by many factors, such as the location and size of the tumor, the presence of extraocular tumor extension, the visual potential of the patient, and the presence of distant metastasis (DM). Treatment options for localized UM include observation, particularly for small and asymptomatic tumors, and local treatments. Radiotherapy (RT) is a highly preferred local treatment of choice as it not only provides a high disease control rate but also aims to preserve the eye and maintain the vision.

Although charged particle beam RT (PBRT) and episcleral plaque brachytherapy (EPB) are the most frequent RT techniques, linear accelerator-based stereotactic radiosurgery and fractionated stereotactic radiotherapy (SRS/FSRT) have been increasingly used in the recent years based on their increasing widespread availability, relatively low cost and noninvasiveness [2,3]. Compared to PBRT and EPB, the data in the literature on SRS/FSRT for the treatment of UM is limited. The present study aims to report the long-term oncological and functional outcomes of SRS/FSRT in patients with UM.

Materials and methods

Patients

The data of patients with UM treated between 2007 and 2019 in our department were retrospectively reviewed. Detailed ophthalmological examination, diagnosis and staging procedures were performed as we described in our previous report [4]. Patients with DM at the time of diagnosis were excluded from the study. This study was approved by the institutional ethics committee and performed in accordance with the Helsinki Declaration.

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Treatment and follow-up

The CyberKnife® (CK®) (Accuray, Sunnyvale, CA, USA) system was used for SRS/FSRT following a peribulbar local anesthetic injection. Details of head and eye immobilization, the procedure of simulation computed tomography (CT), contouring of the target and organs at risk, treatment planning, dose delivery process and follow-up procedure were also described previously [4]. Early and late toxicity were assessed at every follow-up visit.

Statistical analysis

All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS) software version 21 (SPSS Inc., Chicago, IL, USA). The level of statistically significant was set at $p < 0.05$.

Primary endpoints were local control (LC), local recurrence-free survival (LRFS), enucleation-free survival (EFS) and treatment toxicity. Secondary endpoints included overall survival (OS), disease-free survival (DFS), distant metastasis-free survival (DMFS), and eye retention rate. LC was defined as the lack of tumor progression, i.e., any increase in the tumor volume. Complete response was defined as the total regression of the tumor, partial response as a >50% decrease in the tumor volume, and stable disease as all other instances. All time-related events were calculated from the last day of SRS/FSRT to the last follow-up or death. Survival analyses were performed using the Kaplan-Meier method and the results were compared using the log-rank test. Our previous study demonstrated improved outcomes with a total dose of ≥45 Gy [4]. Therefore, in this study, we defined the threshold value of 112.5 Gy for BED_{10Gy} (corresponding 15 Gy in three fractions) for dose comparisons. The rates of LC, eye retention, visual acuity (VA) and treatment toxicity were calculated over the total number of eyes whereas all survival rates were calculated over the number of patients. As peer studies collectively analyzed small and medium tumors, grouping for size-comparing analysis was accomplished as small/medium vs large tumors.

The Cox proportional hazards model was used for multivariate analysis. Factors with a significant contribution to survival estimation ($p < 0.2$) were preserved in the final multivariate model. The hazard ratio (HR) with a 95% confidence interval (CI) was calculated.

Toxicities were graded according to the Common Terminology Criteria for Adverse Events (CTCAE) v4.03. Acute toxicity and late toxicity were defined as toxicity that developed during or within three months and later than three months after the end of SRS/FSRT, respectively.

Results

Four hundred and forty-three patients with 445 UM met the inclusion criteria. Detailed patients, tumor and treatment characteristics are listed in Table 1. The most common SRS/FSRT schedule was 60 Gy in three fractions. One hundred and sixteen (26%) eyes received <45 Gy to the gross tumor volume (GTV) and 329 (74%) tumors received ≥45 Gy. The biologically effective dose when $\alpha/\beta = 10$ (BED_{10Gy}) was ≥180 Gy for 47% of the eyes and the median BED_{10Gy} was 151 (range: 50–211). In all, 66% of the large tumors and 80% of the small/medium tumors received ≥45 Gy ($p = 0.002$). The SRS/FSRT dose and BED_{10Gy} were significantly lower for large tumors compared to small/medium tumors ($p = 0.04$ and $p = 0.03$, respectively).

Table 1
Patient, tumor and treatment characteristics.

Characteristic	Median (range) or n (%)
Age (years)	56 (18–80)
Age group	
≤60 years	287 (65)
>60 years	156 (35)
Sex	
male	231 (52)
female	212 (48)
Tumor site	
Right	216 (49)
Left	225 (51)
Bilateral	2 (0.4)
Visual Acuity (for eyes with available data; n = 386)	0.4
>0.5	144 (37%)
vision loss	223(57)
no light perception	20 (5%)
Tumor diameter (mm)	8 (6–11)
Tumor thickness (mm)	8 (5–11)
COMS classification	
small	44 (10)
medium	267 (60)
large	134 (30)
T stage	
T1	89 (20)
T2	160 (36)
T3	149 (34)
T4	23 (5)
indeterminate	24 (5)
Tumor localization	
posterior	380 (85)
anterior	65 (15)
Baseline ophthalmic examination	
tumor located in and/or invaded the ciliary body	88 (20)
retinal detachment	92 (21)
subretinal hemorrhage	127 (29)
Total dose (Gy)	54 (20–66)
Fraction dose (Gy)	20 (8–22)
Number of fractions	3 (1–5)
Dose-fractionation scheme	
20 Gy in 3 fractions	207 (47)
30 Gy in 3 fractions	68 (15)
54 Gy in 3 fractions	63 (14)
45 Gy in 3 fractions	38 (9)
20 Gy in 1 fraction	21 (5)
Others	48 (10)
Treatment days (for FSRT; n = 396)	
consecutive	269 (68)
every other day	127 (32)
BED_{3Gy}	378 (66–550)
BED_{10Gy}	151 (50–211)
BED_{10Gy} group	
<112.5	115 (25)
≥112.5	328 (75)
CI	1.29 (1.03–2.57)
HI	1.16 (1.09–1.50)
Collimator size (mm)	5 (5–20)
Normalization (%)	86 (67–151)
Number of beams	173 (87–295)
Size of GTV (mm^3)	603 (23–3,808)
Maximum dose to the GTV (Gy)	65.9 (25.7–80.2)
Maximum dose to the ipsilateral eye (Gy)	65.8 (25–80.2)
Maximum dose to the contralateral eye (Gy)	4.4 (0.4–9.8)
Maximum dose to the ipsilateral lens (Gy)	15.3 (5.2–80.1)
Maximum dose to the contralateral lens (Gy)	1.9 (0.1–5.2)
Maximum dose to the ipsilateral optic nerve (Gy)	20.6 (9.6–50.2)
Maximum dose to the contralateral optic nerve (Gy)	2.8 (0.6–8)
Maximum dose to the optic chiasm (Gy)	1.86 (0.12–9.3

Abbreviations: IQR = inter-quartile range, COMS = Collaborative Ocular Melanoma Study, FSRT = fractionated stereotactic radiotherapy, BED_{3Gy} = biologically effective dose when $\alpha/\beta = 3$, BED_{10Gy} = biologically effective dose when $\alpha/\beta = 10$, CI = conformity index, HI = homogeneity index, GTV = gross tumor volume.
*BED calculated with the formula; dose/fraction × fraction number (1 + fraction dose/α/β).

Median follow-up was 74 months (range: 1–173 months) in all patients. The overall LC rate was 83%. The LC rate was significantly higher in eyes with small/medium tumors (88% vs 70% for large tumors, $p < 0.001$) and irradiated with ≥45 Gy (88% vs 69% for <45 Gy, $p < 0.001$) (Fig. 1a and 1b). Patient age, gender and treatment day schedule for FSRT did not impact the LC rate. The 2-, 5-, and 10-year LRFS rate was 88%, 74%, and 56%, respectively. Median LRFS was not reached at the time of analysis. In univariate analysis, younger age, a smaller tumor size, a higher total SRS/FSRT dose and a higher BED_{10Gy} value resulted in a significantly higher LRFS rate (Table 2). Following a total dose of ≥45 Gy, the 2-, 5- and 10-year LRFS rate was 91%, 89% and 83% for small/medium tumors, and 86%, 63% and 46% for large tumors, respectively ($p = 0.002$). In multivariate analysis, patient age, tumor size, SRS/FSRT dose and BED_{10Gy} were independent prognostic factors for LRFS (Table 3). The LC rate did not differ according to the treatment day schedule (every other day vs consecutive) ($p = 0.08$).

One hundred and sixty-six (38%) eyes were enucleated during the follow-up. The overall eye retention rate was 62%, and it was significantly different with regard to the tumor size (73% for small/medium vs 36% for large tumors, $p < 0.001$), total SRS/FSRT dose (69% for ≥45 Gy vs 31% for <45 Gy, $p < 0.001$), and treatment day schedule for FSRT (81% for every other day vs 54% for consecutive days, $p < 0.001$). Analysis of the relationship between the tumor size and SRS/FSRT dose showed that 73% of large tumors receiving <45 Gy and 59% receiving ≥45 Gy underwent enucleation ($p = 0.13$) whereas in small/medium tumors, the respective rates were 49% and 21% ($p < 0.001$). There was a significant difference in the enucleation rate between large and small/medium tumors receiving <45 Gy (73% vs 49%, $p = 0.025$) and ≥45 Gy (59% vs 21%, $p < 0.001$). With regard to the treatment day schedule, consecutive treatments resulted in an enucleation rate of 34% in small/medium and 65% in large tumors ($p = 0.001$) whereas the respective rates were 15% and 41% for every-other-day treatments ($p = 0.065$). In addition, although the treatment day schedule did not significantly affect the enucleation rate for tumors receiving <45 Gy (65% for consecutive vs 64% for every-other-day, $p = 1$), treatment on consecutive days significantly increased the rate of enucleation for tumors receiving ≥45 Gy (40% vs 13% for every-other-day treatment, $p < 0.001$).

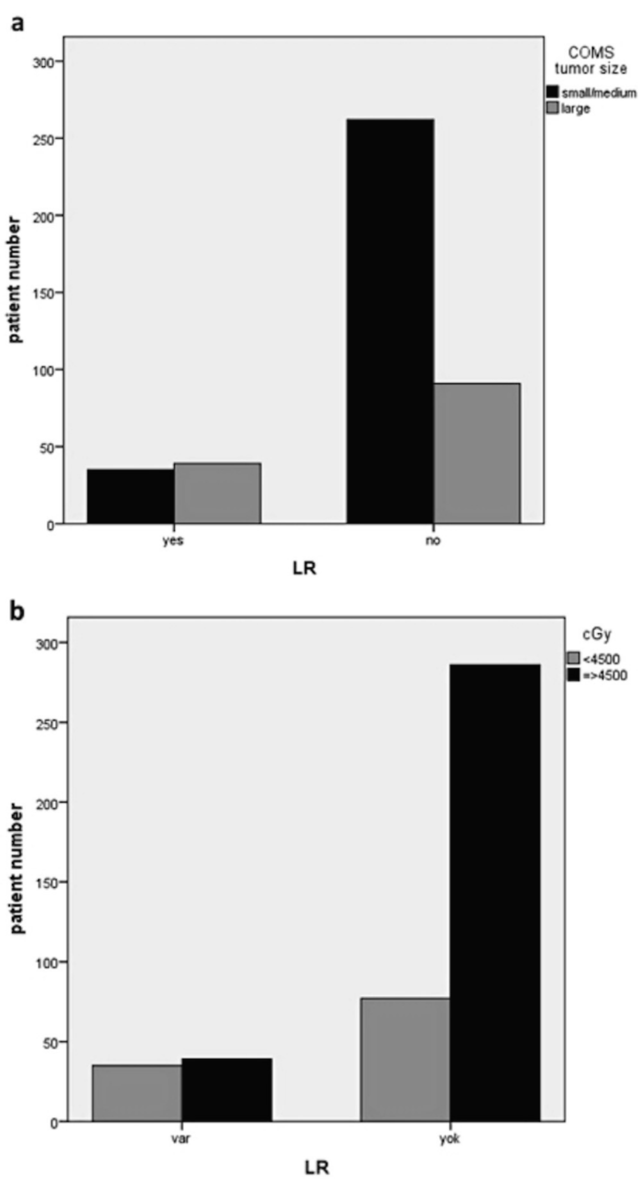

Fig. 1. Rates of Local Recurrence for Tumor Size (1a) and SRS/FSRT Dose (1b).

Of the 166 enucleated eyes, the reason for enucleation was tumor progression in 98 (59%) and treatment-related complications in 58 (35%) eyes, respectively. The reason for enucleation could not be specified in 10 eyes as these patients underwent surgery at other facilities. For the 156 eyes in which the indication for enucleation could be specified, the indication did not differ between large and small/medium tumors (progression in 58% vs 71%, and complication in 43% vs 29%, $p = 0.09$) or for ≥45 Gy and <45 Gy (progression in 57% vs 71%, and complication in 43% vs 30%, $p = 0.13$). However, treatment on consecutive days resulted in a higher rate of complication-related enucleation compared to treatment every other day (42% vs 19%, and progression in 58% vs 81%, $p = 0.048$).

The 2-, 5- and 10-year EFS rate was 83%, 64%, and 36%, respectively, and median EFS was 88 months (standard error [SE]: 5.27, 95% CI: 78.04–98.69). The EFS rate was significantly higher in small/medium tumors, with ≥45 Gy, and with a BED_{10Gy} of ≥112.5 Gy (Table 2). Following ≥45 Gy, the 2-, 5- and 10-year EFS rate was 88%, 75%, and 52% for small/medium tumors and 79%, 57%, and 24% for large tumors respectively ($p < 0.001$). Tumor size, total SRS/FSRT dose, and BED_{10Gy} were independent prognostic factors for EFS in multivariate analysis (Table 3).

The 2-, 5-, and 10-year OS rate was 97%, 87%, and 71%, respectively, and median OS was not reached at the time of analysis. In univariate analysis, a higher OS rate was associated with a younger age and a smaller tumor size (Table 4). Following ≥45 Gy, the 2-, 5-, and 10-year OS rate was 95%, 89%, and 77% for small/medium tumors and 98%, 81%, and 64% for large tumors, respectively ($p = 0.097$). The multivariate analysis revealed the patient age and COMS size as independent prognostic factors for OS (Table 5). Local control was not associated with OS ($p = 0.2$).

The 2-, 5- and 10-year DFS rate was 87%, 74% and 56%, respectively. The DFS rate significantly decreased with an increased tumor size, decreased total SRS/FSRT dose, decreased BED_{10Gy} and older age (Table 4). Following ≥45 Gy, the 2-, 5- and 10-year DFS rate was 91%, 83%, and 70% for small/medium tumors and 84%, 66%, and 48% for large tumors, respectively ($p = 0.001$). In multivariate analysis, tumor size, patient age, SRS/FSRT dose and BED_{10Gy} were independent prognostic factors for DFS (Table 5).

DM developed in 54 (12%) patients. The most common site of DM was the liver, followed by the lung and bones. The 2-, 5- and 10-year DMFS rate was 93%, 82%, and 70%, respectively. In univariate analysis, patient age and COMS tumor size were significant factors for DMFS (Table 4). Following ≥45 Gy, the 2-, 5- and 10-year DMFS rate was 94%, 85%, and 77% for small/medium tumors, and 92%, 74%, and 61% for large tumors, respectively ($p = 0.02$). Patient age and COMS tumor size were independent prognostic factors for DMFS in multivariate analysis (Table 5).

Table 2
Results of Univariate Analysis for LRFS and EFS.

Survival and factor	2y LRFS (%)	5y LRFS (%)	10y LRFS (%)	p	2y EFS (%)	5y EFS (%)	10y EFS (%)	p
Age								
≤60 years	88	76	63	0.009	82	64	42	0.1
>60 years	90	70	44		86	64	24	
Gender								
Male	86	72	54	0.4	78	60	36	0.4
Female	91	76	59		89	69	35	
COMS size								
Small/medium	90	79	66	<0.001	86	70	50	<0.001
Large	85	64	42		76	52	20	
SRS/FSRT dose								
<45 Gy	85	63	46	0.002	77	50	28	0.002
≥45 Gy	90	79	61		86	70	39	
BED_{10Gy} group								
<112.5	78	62	44	0.004	79	52	29	0.004
≥112.5	90	78	59		86	70	38	
Treatment days*								
Consecutive	89	74	57	0.9	81	63	35	0.3
Every other day	87	81	41		86	70	21	

Abbreviations: y = year, LRFS = local recurrence-free survival, EFS = enucleation-free survival, COMS = Collaborative Ocular Melanoma Study, SRS/FSRT = stereotactic radiosurgery/fractionated stereotactic radiotherapy, BED = biologically effective dose.
* For FSRT.

Table 3
Results of Multivariate Analysis for LRFS and EFS.

	HR	95% CI	p-value
LRFS			
Age			
≤60 years	1	1.1–2.15	0.012
>60 years	1.54		
COMS tumor size			
Small/medium	1	1.26–2.46	0.001
Large	1.76		
SRS/FSRT dose			
≥45 Gy	1	1.25–2.49	0.001
<45 Gy	1.77		
BED_{10Gy} group			
<112.5	1	0.41–0.81	0.002
≥112.5	0.58		
EFS			
Age			
≤60 years	1	0.92–1.75	0.06
>60 years	1.3		
COMS tumor size			
Small/medium	1	1.37–2.39	<0.001
Large	1.81		
SRS/FSRT dose			
≥45 Gy	1	1.17–2.11	0.003
<45 Gy	1.57		
BED_{10Gy} group			
<112.5	1	0.48–0.89	0.007
≥112.5	0.65		

Abbreviations: LRFS = local recurrence-free survival, COMS = Collaborative Ocular Melanoma Study, SRS/FSRT = stereotactic radiosurgery/fractionated stereotactic radiotherapy, BED = biologically effective dose, EFS = enucleation-free survival, HR = hazard ratio, CI = confidence interval.

The treatment was well tolerated with no grade ≥3 treatment-related acute toxicity. The risk of development of acute toxicity was not associated with the total SRS/FSRT dose, tumor size or treatment day schedule (p = 0.2, p = 0.3, and p = 0.1, respectively).

Late toxicity was observed in 207 (47%) eyes. The most common late toxicity was radiation retinopathy followed by optic neuropathy, cataracts, neovascular glaucoma, vitreous hemorrhage, and retinal detachment. Radiation retinopathy developed in 161 (36%) eyes but the mean eye dose was not associated with the development of this toxicity (p = 0.3). Optic neuropathy occurred in 120 (27%) eyes and a higher maximum optic nerve dose significantly increased the rate of neuropathy (p = 0.03). Radiation-induced cataracts developed in 79 (18%) eyes. The mean lens dose was significantly higher in the eyes with radiation-induced cataracts than those without (25 Gy vs 14 Gy, p = 0.008). The rate of late toxicity was higher in eyes with a large tumor compared to the eyes with a small/medium tumor (p = 0.04) and also in the high-dose SRS/FSRT group than in the low-dose group (p = 0.003).

Among the 396 eyes treated with FSRT, the treatment day schedule was a significant prognostic factor for the development of toxicity. The late toxicity rate was significantly higher in the eyes treated on consecutive days than those treated every other day (52% vs 38%, p = 0.009). In 75% of eyes with late toxicity, FSRT was delivered on consecutive days.

At the last follow-up, the median VA was 0.1 and significantly worse than the basal median VA (p = 0.02). The VA deteriorated in 185 (76%) eyes, was stable in 54 (22%) eyes, and improved in 5 (2%) eyes following SRS/FSRT. No relation was found between the VA change and the total dose, tumor size or treatment day schedule.

Discussion

In the current study which is the largest SRS/FSRT series for the treatment of UM in the literature, we report the 5- and 10-year LRFS rate 74% and 56%, respectively, and the overall LC rate 83% after a median 74-month follow-up. Eyes with small/medium tumors and those irradiated with a ≥45 Gy total dose had significantly higher survival rates. For all eyes treated with ≥45 Gy, all survival rates were significantly lower for large tumors compared to small/medium tumors, mainly due to the higher rate of DM. On the contrary to the satisfying LC rate, SRS/FSRT was not without disappointment. The treatment-related late toxicity was observed in nearly half of the eyes and lead to the formidable enucleation. The eye retention rate was significantly higher and toxicity rates lower for small/medium tumors than for large tumors. The most striking finding of this study is the significant decrease in the rate of late toxicity in eyes treated every other day compared to the ones treated on consecutive days with FSRT without compromising oncologic outcomes. Consecutive treatment led to a higher enucleation rate for large tumors irradiated with ≥45 Gy.

Surgery was largely replaced by EPB and PBRT based on retrospective evidences with comparable results with the aim to preserve the eye and maintain vision [5–9]. However, the insufficient target coverage in large tumors, invasive intervention,

Table 4
Results of Univariate Analysis for OS, DFS and DMFS.

Survival and factor	2y OS (%)	5y OS (%)	10y OS (%)	p	2y DFS (%)	5y DFS (%)	10y DFS (%)	p	2y DMFS (%)	5y DMFS (%)	10y DMFS (%)	p
Age												
≤60 years	98	90	78	<0.001	87	76	62	0.012	94	85	76	<0.001
>60 years	94	80	57		88	68	44		92	77	57	
Gender												
Male	95	84	71	0.5	85	71	53	0.23	92	79	69	0.4
Female	98	90	72		90	76	60		95	85	71	
COMS size												
Small/Medium	96	89	78	0.02	89	78	65	<0.001	94	85	77	0.002
Large	98	82	63		81	63	42		91	75	60	
SRS/FSRT dose												
<45 Gy	98	86	72	0.8	82	63	46	0.002	92	82	70	0.9
≥45 Gy	96	87	71		89	78	61		94	82	70	
BED$_{10Gy}$ group												
<112.5	98	86	72	0.7	83	62	44	0.003	94	84	73	0.9
≥112.5	96	86	70		88	76	60		94	85	71	
Treatment days[*]												
Consecutive	97	86	71	0.6	87	74	57	0.83	94	82	69	0.6
Every other day	95	90	82		87	80	41		92	82	82	

Abbreviations: y = year, OS = overall survival, DFS = disease-free survival, DMFS = distant metastasis-free survival, COMS = Collaborative Ocular Melanoma Study, SRS/FSRT = stereotactic radiosurgery/fractionated stereotactic radiotherapy, BED = biologically effective dose.
[*] For patients irradiated in more than one fraction.

Table 5
Results of Multivariate Analysis for OS, DFS and DMFS.

	HR	95% CI	p-value
OS			
Age			
≤60 years	1	1.46–3.32	<0.001
>60 years	2.2		
COMS tumor size			
Small/medium	1	1.08–2.48	0.021
Large	1.6		
DFS			
Age			
≤60 years	1	1.08–2.09	<0.015
>60 years	1.51		
COMS tumor size			
Small/medium	1	1.29–2.5	0.001
Large	1.79		
SRS/FSRT dose			
≥45 Gy	1	1.25–2.48	0.001
<45 Gy	1.76		
BED$_{10Gy}$ group			
<112.5	1	0.42–0.82	0.002
≥112.5	0.59		
DMFS			
Age			
≤60 years	1	1.28–2.79	0.001
>60 years	1.89		
COMS tumor size			
Small/medium	1	1.25–2.73	0.002
Large	1.85		

Abbreviations: HR = hazard ratio, CI = confidence interval, OS = overall survival, COMS = Collaborative Ocular Melanoma Study, DFS = disease-free survival, SRS/FSRT = stereotactic radiosurgery/fractionated stereotactic radiotherapy, BED = biologically effective dose, DMFS = distant metastasis-free survival.

technical impossibility in lesions around the optic nerve and high radioactive source costs of EPB, and the requirement of sophisticated equipment for PBRT available only at selected centers have put SRS/FSRT forward in the recent years [3,10,11]. Nevertheless, experience is even more limited with SRS/FSRT and there are no trials directly comparing its results with either of the other techniques. SRS/FSRT has important advantages over other techniques such as a higher dose conformity, widespread availability, relatively low cost and non-invasiveness. Results from single-center series suggest that SRS/FSRT can be as effective as other RT forms only with a slightly higher risk of toxicity [11–15].

Studies with definitive SRS/FSRT for UM have reported high LC rates between 80% and 98% [2,3,10,12–28]. Although the overall LC rate of 83% in the current study is within this range, the high rate of large tumors may have resulted in a lower-than-expected rate. Recently, Eibenberger et al. [29] reported the long-term results of SRS in 335 patients with large tumors. The median tumor thickness was 4.6 mm and the 10-year LC rate was reported 95%. This higher LC rate can be attributed to the smaller median tumor size in that study. In another study including 24 patients treated with photon SRS, the 5-year LRFS rate was reported 82% with a median LRFS of 66 months which is not reached in our study yet after a median 74-month follow-up [30]. With a median thickness of 8 mm and longer than a 10-year median follow-up in 445 eyes, we think that the 83% overall LC rate in our study is highly satisfying.

The most commonly used RT techniques for the treatment of UM are EPB and PBRT which have comparable LC rates [31,32]. Based on retrospective studies, SRT can be reported to have similar outcomes compared to both techniques [33]. However, no randomized study has compared their efficacy and the best treatment is still unknown. There are studies of EPB with comparable LC rates to ours; however, both the number of patients and tumor size are smaller than the ones in our study [34,35]. On the other hand, the COMS report No.19 reported a LC rate of 89.7% with ^{125}I EPB [36]. Besides, LC control rates of up to 95% at five years have been reported with proton therapy. However, in a study of medium and large tumors treated with PBRT, the 5-year LC rate was reported to decrease with increasing tumor diameter and was found 71% for tumors >16 mm [37]. Albeit our results seem inferior to EPB and PBRT, it should be kept in mind that 30% of our patients had large tumors. In addition, comparing treatment outcomes from different studies is not optimal because of confounding factors such as patient selection criteria, tumor characteristics, and definition of the recurrence rate of the studies.

Various potential prognostic factors have been investigated for UM and some of these are still debated. Since there is no randomized study investigating the relationship between the dose and oncologic outcomes, this issue is still controversial. In a prospective study from Harvard University investigating the dose effect for small/medium tumors, 50 GyE and 60/70GyE proton in five fractions were compared [29]. No difference was observed between the two groups regarding LC and eye retention rates. Similarly, in a retrospective study including 190 patients, increasing doses with

^{125}I PBT did not improve oncologic outcomes but increased the toxicity rates [38]. In a systematic review which included more than 2,000 patients treated with EPB in 16 studies, local recurrence improved for each 1-Gy increase although not statistically significant [39]. We observed improved LRFS, EFS and DFS in eyes that received $\geq$45 Gy compared to those that received <45 Gy. Similar to the review mentioned above, we could not find an impact of dose on OS or DMFS.

All forms of RT can be associated with ocular toxicity, some being more specific to the RT technique. The overall toxicity rate and most common toxicities in our study are similar to the previous reports [2,3,11,12,19,24,29,40]. Although SRS/FSRT was well tolerated in the acute phase, half of our patients suffered late toxicity. Some studies have linked the toxicity rate to the RT dose, tumor localization, and tumor size [19,21,24,41–43]. In our study, higher rates of cataracts and neovascular glaucoma were observed in anteriorly-located tumors but the VA was deteriorated more in posteriorly-located tumors. Moreover, a higher rate of toxicity was observed in the eyes treated on consecutive days compared to those treated every other day. The median VA of our cohort at diagnosis was 0.4 and significantly declined to 0.1 in the follow-up. A similar decline was previously reported by other series [2,3,24,29]. After SRS/FSRT, the eye retention rates were reported 73–90% [3,4,14,20–24,26,44–46]. In our study, the overall eye retention rate was 62% and it was higher for eyes with small/medium tumors and for eyes receiving $\geq$45 Gy. The enucleation rate as well as the toxicity-related enucleation rate were significantly lower in patients that received the treatment every other day. It has been shown that every other day treatment resulted in decreased toxicity in FSRT for various sites [47,48]. To the best of our knowledge, there is no study analyzing the effect of interval between fractions for FSRT in patients with UM. This makes our study remarkable.

Our results should be cautiously evaluated due to its limitations. The major limitation of our study is its retrospective design which resulted in incomplete data on patient and tumor characteristics and treatment results. Additionally, our study focused on SRS/FSRT outcomes per se and did not compare it to enucleation or other RT forms. Randomized studies should compare these modalities to determine whether SRS/FSRT yields similar outcomes with other RT forms.

Conclusion

Our study is the largest series in the literature using SRS/FSRT for the treatment of UM. A $\geq$45 Gy total dose of SRS/FSRT and BED_{10Gy} is associated with higher LC, even in eyes with COMS large tumors. However, despite the favorable oncologic outcomes, SRS/FSRT-related toxicity still represents a serious limitation. Herein, we report that toxicity can be minimized by treating the eye every other day without compromising oncologic outcomes.

Conflict of Interest

Any actual or potential conflicts of interest do not exist.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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